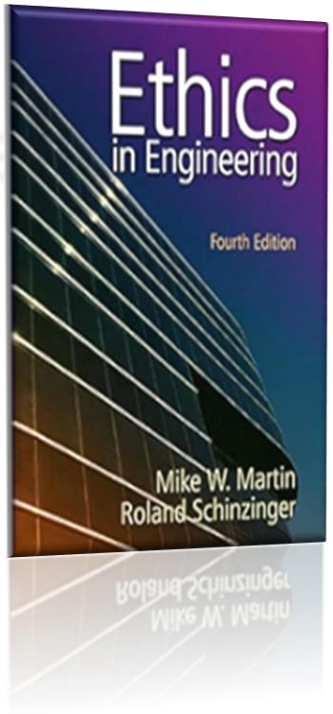
Engineering Ethics (HSS-422) (3+0)



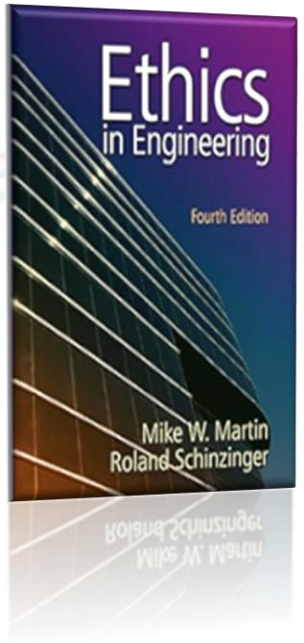




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COMMUNICATING RISK AND PUBLIC POLICY

Week No 11

By Engr. Syed Rizwan Ali

Learning Outcomes

In this Lecture will cover ..

- Understanding Risk Identification Techniques:
- To Assess and Prioritize Risks:
- To Develop Risk Mitigation Strategies:
- To understand Legal and Ethical Responsibilities:
- To Engage with Stakeholders:

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COMMUNICATING RISK AND PUBLIC POLICY



Communicating Risk And Public Policy

❖ **Background:**

- Communicating risk and public policy involves effectively conveying information about potential hazards and the measures taken to mitigate them.
- This process is critical in various fields, including engineering, public health, environmental safety, and more.
- The primary goal is to ensure that all stakeholders, including the public, understand the risks and the policies in place to manage them.

Communicating Risk and Public Policy

- **Key Components of Communicating Risk and Public Policy are as follows:**
- Clarity and Accuracy:
- Understanding the Audience:
- Transparency:
- Engagement:
- Multiple Channels:
- & Feedback Mechanisms:

Communicating Risk and Public Policy

- **Key Components of Communicating Risk and Public Policy are as follows:**
- Clarity and Accuracy: Information must be presented clearly and accurately to avoid misunderstandings. It's essential that the data is up-to-date and based on sound scientific evidence.
- Understanding the Audience: Different audiences may require different approaches depending on their level of knowledge, cultural background,

Communicating Risk and Public Policy

- **Key Components of Communicating Risk and Public Policy are as follows:**
- Transparency: Being transparent about uncertainties and the limits of current knowledge helps build trust. Communicators should not only share what is known, but also what is not known.
- Engagement: Interactive communication strategies that involve the audience can enhance understanding and retention of information. This might include public forums, interactive web tools, or community meetings.

Communicating Risk and Public Policy

- **Key Components of Communicating Risk and Public Policy are as follows:**
- Multiple Channels: Utilizing a variety of communication channels (e.g., social media, news outlets, direct community engagement) ensures broader reach and accessibility of the information.
- Feedback Mechanisms: Establishing ways for the audience to provide feedback or ask questions can improve ongoing communication efforts and help address public concerns more effectively.

Difficulties in Determining Causes & Likelihood of Harm

- **Background:**

- Determining the causes and likelihood of harm in risk assessment and public policy can be challenging for several reasons.
- These difficulties affect how risks are communicated and managed in various fields, from public health to environmental safety. Here are some key challenges:.

Complex Causality	Changing Conditions
Data Limitations	Interdisciplinary Challenges
Scientific Uncertainty	Psychological Factors
Long Latency Periods	Ethical and Political Influences

Difficulties in Determining Causes & Likelihood of Harm

- **Complex Causality:** Many risks involve complex interactions between various factors, making it difficult to isolate a single cause.
- For instance, environmental health risks may result from a combination of exposure to pollutants, genetic factors, and lifestyle choices.
- **Data Limitations:** There might be limited data on certain risks, especially new or emerging ones. Lack of historical data can make it hard to predict future occurrences or impacts accurately.

Difficulties in Determining Causes & Likelihood of Harm

- **Scientific Uncertainty:** Science often deals with probabilities and uncertainties. Communicating these uncertainties to non-experts without causing unnecessary alarm or skepticism is a significant challenge.
- **Changing Conditions:** The likelihood of certain risks can change over time due to factors like climate change, technological advances, or changes in social behavior, which can make previous assessments outdated.

Difficulties in Determining Causes & Likelihood of Harm

- **Interdisciplinary Challenges:** Effective risk assessment often requires integrating knowledge from various disciplines, which can be difficult due to differing methodologies, terminologies, and priorities.
- **Psychological Factors:** People's perceptions of risk can be influenced by emotional, cognitive, and social factors, which may not align with the actual risk based on scientific evidence. This can affect public responses to risk communication.

Difficulties in Determining Causes & Likelihood of Harm

- **Ethical and Political Influences:** Decisions about risk management can be influenced by ethical, political, and economic considerations, which may complicate the objective assessment of the causes and likelihood of harm.
- These challenges highlight the need for robust methodologies in risk assessment and innovative approaches to communicate risks effectively, ensuring that policies are both scientifically sound and publicly accepted.

The Critical Attitude - Engineer's Liability

❖ **Background:**

- The concept of an engineer's liability intertwined with a critical attitude towards their work is fundamental in engineering ethics and professional responsibility.
- This perspective emphasizes the importance of meticulousness, foresight, and a proactive stance in addressing potential risks and ethical dilemmas in engineering projects. Here are key aspects of this approach:

The Critical Attitude - Engineer's Liability

- **1. Professional Competence:**
- Engineers are expected to maintain a high level of technical and professional competence. This involves continual learning and staying updated with the latest advancements and standards in their field.
- Competence is the foundation upon which trust in engineering judgments is built, directly impacting the engineer's liability for their work.

The Critical Attitude - Engineer's Liability

- **2. Critical Evaluation**
- Engineers must critically evaluate all aspects of their projects, from design to implementation. This includes assessing potential risks, the environmental impact, and the long-term sustainability of the project.
- A critical attitude helps identify possible failures or shortcomings before they manifest, thus reducing liability risks.

The Critical Attitude - Engineer's Liability

- **3. Ethical Decision-Making:**
- Engineering decisions often involve trade-offs and ethical considerations, such as balancing cost with safety or environmental impact.
- Engineers must adhere to a strict code of ethics, prioritizing public safety, health, and welfare above all else. Ethical lapses can lead to significant legal and professional liabilities.

The Critical Attitude - Engineer's Liability

- **4. Risk Management:**
- A critical component of reducing liability is effective risk management. Engineers need to identify, analyze, and mitigate risks associated with their projects.
- This includes preparing for unforeseen issues through robust contingency planning

The Critical Attitude - Engineer's Liability

- **5. Communication:**
- Effective communication with stakeholders, clients, and the public is crucial. Engineers must clearly explain the technical aspects, risks, and ethical considerations of their projects.
- Transparency helps build trust and reduces misunderstandings that could lead to liability issues.

The Critical Attitude - Engineer's Liability

- **6. Documentation:**
- Meticulous documentation of all phases of a project is essential. This serves as a record of the decisions made, the basis for these decisions, and compliance with relevant standards and regulations.
- In the event of a dispute or legal challenge, comprehensive documentation can significantly protect against liability.

The Critical Attitude - Engineer's Liability

- **7. Accountability:**
- Engineers must accept responsibility for their work and its outcomes. This accountability is a central aspect of professional ethics and liability. Acknowledging and learning from mistakes, rather than concealing them, is critical in fostering a culture of safety and integrity.
- By cultivating a critical attitude and rigorously adhering to ethical and professional standards, engineers can effectively manage their liability and contribute positively to their field.

Case Study: Hyatt Regency Walkway

- **Case:**
- A real-world example that illustrates the critical attitude and liability of engineers can be found in the case of the Hyatt Regency walkway collapse in 1981.
- This disaster is one of the most studied failures in the field of engineering and offers profound lessons on the importance of ethical responsibility and technical diligence.

Case Study: Hyatt Regency Walkway

- **Background:**
- The Hyatt Regency Hotel in Kansas City, Missouri, featured a unique design with suspended walkways crossing the atrium at several levels. On July 17, 1981, during a tea dance, the fourth-floor walkway collapsed onto the second-floor walkway below, which then fell into the crowded lobby. The incident resulted in 114 deaths and over 200 injuries, making it one of the deadliest structural collapses in American history.

Case Study: Hyatt Regency Walkway

- **Engineering Failures and Liabilities:**
- The investigation revealed multiple critical issues related to engineering decisions:
- **Design Changes:** The original design of the walkways featured a single rod supporting both the fourth-floor walkway and the second-floor walkway directly beneath it. However, during construction, the design was modified such that two separate sets of rods were used for each walkway.

Case Study: Hyatt Regency Walkway

- **Engineering Failures and Liabilities:**
- The investigation revealed multiple critical issues related to engineering decisions:
- **Design Changes:** This change doubled the load on the fourth-floor walkway's connector, a factor not properly evaluated by the responsible engineers.
- **Lack of Oversight:** The engineering firm responsible for the design failed to review the changes adequately.

Case Study: Hyatt Regency Walkway

- **Engineering Failures and Liabilities:**
- **Lack of Oversight:** The modifications were communicated through shop drawings and not subjected to the rigorous scrutiny typically required for such critical structural elements.
- **Communication Breakdown:** Significant lack of communication between the engineering team and the construction contractors. The critical nature of the design change was not understood by all parties, and the implications of the modification were not fully assessed.

Case Study: Hyatt Regency Walkway

- **Legal and Ethical Consequences:**
- The disaster led to significant legal consequences for the engineering firm involved. The engineers who approved the final drawings were found negligent and liable for the collapse. Their professional licenses were revoked, and they faced substantial financial penalties. This case underscored the liability engineers hold for ensuring their designs meet safety standards and for rigorously checking and approving any changes to their plans.

Case Study: Hyatt Regency Walkway

- **Lessons Learned:**
- The Hyatt Regency disaster serves as a stark reminder of the critical role engineers play in public safety and the severe consequences of negligence.
- Rigorous adherence to safety standards.
- Thorough review and approval processes for design changes.
- Effective communication among all parties involved in a project.
- Ethical responsibility to prioritize safety over cost or aesthetic considerations

Responsible Engineer Regarding Risk

- **Background:**
- Becoming a responsible engineer, especially in the context of managing and communicating risk, involves several key practices and attitudes.
- Here are some essential bullet points for engineers aiming to be responsible in their handling of risks:
- **Understand and Assess Risks**
- **Commit to Continuous Learning**
- **Adhere to Safety Standards and Regulations Implement Risk Management Plans**

Responsible Engineer Regarding Risk

- **Background:**
- **Communicate Risks Effectively**
- **Foster a Culture of Safety**
- **Practice Ethical Decision-Making**
- **Document All Processes**
- **Be Proactive, Not Just Reactive**
- **And Take Responsibility for Your Work**
- These points underscore the importance of a holistic and proactive approach to risk management in engineering, ensuring that safety and ethical considerations are at the forefront of every project.

The Engineer's Liability for Risk Becoming a Responsible Engineer Regarding Risk

- **Uphold Professional Standards:** Adhere strictly to the engineering codes of ethics and professional standards. This compliance helps minimize legal liability and ensures public safety.
- **Conduct Thorough Risk Assessments:** Regularly perform comprehensive risk assessments for all projects. Identify potential hazards and evaluate their impact to effectively manage risks.
- **Maintain Competence:** Continuously update your knowledge and skills related to your engineering field, including the latest risk management techniques and technologies.

The Engineer's Liability for Risk Becoming a Responsible Engineer Regarding Risk

- **Communicate Risks Clearly:** Effectively communicate potential risks to all stakeholders, ensuring that they understand the implications and the steps being taken to mitigate these risks.
- **Document Decisions and Actions:** Records of all decisions and actions related to risk management. Documentation can serve as evidence of diligence and adherence to standards in case of legal scrutiny.
- **Implement Safety Measures:** Design and implement appropriate safety measures that meet or exceed legal requirements. Proactively address any identified risks to prevent accidents or failures.

The Engineer's Liability for Risk Becoming a Responsible Engineer Regarding Risk

- **Transparent with Stakeholders:** Maintain open lines of communication with all stakeholders, including clients, colleagues, and the public, about risk factors and mitigation strategies.
- **Take Responsibility:** Accept accountability for the safety of your designs and decisions. Be prepared to address and rectify any issues that arise from overlooked risks or mistakes.
- **Seek Peer Reviews:** Regularly have your work reviewed by peers to catch potential oversights and improve the safety and reliability of your projects.

The Engineer's Liability for Risk Becoming a Responsible Engineer Regarding Risk

- **Promote a Safety Culture:** Encourage a culture within your team or organization where safety and risk management are prioritized, and where everyone feels responsible and empowered to speak up about risks.

Thanks

Any Question

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